

Single-cell foundation models represent genes in context, but the information is co-expression

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Abstract

A single-cell foundation model reads a cell as a sequence of its genes and produces, at each layer, a contextual representation of every gene — the gene’s vector given the other genes in that cell. Whether that representation carries usable information about a gene’s role in a particular cell is unknown, and is the single-cell analogue of word-sense disambiguation in language models. We give a systematic answer. Across three architectures (MaxToki, scGPT, STATE) a gene’s representation is reshaped by cellular context in a way that is gene-specific (cross-cell agreement 0.759 versus 0.001 for a gene-shuffled null), reproducible across independent cells, not an artefact of the gene’s rank position (96% survives rank control), and replicable on an independent developmental dataset. It is causally used: steering a functional direction into half a cell’s genes shifts the model’s own predictions for the other genes ($p = 7 \times 10^{-5}$). It is learned rather than architectural: an untrained model of the same architecture organises context far less ($z = +3.8$ versus $+21$ for the trained model). It also increases with model size, though this scaling increment is partly confounded by embedding width. The context modulation is organised along interpretable functional axes, but against size-matched co-expression nulls this organisation does not robustly exceed co-expression — the strongest axis is only at the $p = 0.05$ border and the others are non-significant — and two independent probes for novel context-dependent gene function are null. The useful, ceiling-immune output is a model metric: contextualisation and its functional organisation differ sharply by tokenisation (rank-based versus value-binned versus protein-sequence-token) and scale with size. We report the phenomenon, place a boundary on what it encodes, and give the metric.

Keywords: single-cell foundation models; contextual gene representations; mechanistic interpretability; co-expression; gene embeddings; word-sense disambiguation; model characterisation.

1. Introduction

A recurring theme in the interpretability of large language models is that a network trained on a narrow predictive task builds internal representations of structure it was never taught. Sequence models of Othello and chess build representations of the board and of latent variables such as player skill (Li et al. 2023; Nanda et al. 2023; Karvonen 2024); language models linearly encode space and time along identifiable directions (Gurnee and Tegmark 2024); and, in the best cases, these representations are not decorative but *causally used* — a claim one tests by intervening on the representation and measuring the effect on behaviour (Turner et al. 2023; Zou et al. 2023; Park et al. 2024).

A closely related line concerns *contextual* representations. Contextual word embeddings assign the same word a different vector in every sentence, and the geometry of those vectors encodes word sense: “bank” near a river and “bank” in finance occupy separable regions (Peters et al. 2018; Ethayarajh 2019; Coenen et al. 2019). A cottage industry of methodological care surrounds these measurements, because representation spaces are strongly anisotropic (vectors cluster in a narrow cone rather than spreading evenly, so any two look similar) and a handful of “rogue” dimensions

can dominate cosine similarity (Ethayarajh 2019; Mu et al. 2018; Cai et al. 2021; Timkey and Schijndel 2021).

Single-cell foundation models (SCFMs) offer a clean biological version of the same question (Theodoris et al. 2023; Cui et al. 2024; Hao et al. 2024; Yang et al. 2022). They treat each gene as a token and each cell as a sequence of its expressed genes, and are trained by masked or autoregressive prediction over expression-ranked genes — in effect, learning which genes tend to be active together. Their gene *token* embeddings (one fixed vector per gene) have been studied; their *contextual* gene representations (a gene’s vector inside a particular cell) are used as features downstream (Hao et al. 2024; Bian et al. 2024) but have not been characterised as objects in their own right. Yet this is exactly where a model could, in principle, represent that a gene plays a different role in a macrophage than in a T cell — the biological analogue of word sense, and, as W. Gilpin has noted, an open direction that the “cell as a token” framing invites but does not test (W 2025).

We organise the question into two levels. **Level 1 (a model-fact):** is a gene’s representation reshaped by context in a structured, gene-specific way, and is that reshaping learned and used? **Level 2 (biology):** does the reshaping reveal something *biologically novel* about the gene’s role in each context — beyond what co-expression already tells us? We answer Level 1 affirmatively and comprehensively, and Level 2 negatively and with matched-null rigour. The negative is not a failure to report but the paper’s second contribution: it locates, precisely and with controls, the ceiling that a growing critical literature has found for SCFMs in other guises (Kedzierska et al. 2025; Boiarsky et al. 2024; Ahlmann-Eltze et al. 2025; Kendiukhov 2026e). The paper thus makes three contributions: (1) the Level-1 model-fact, characterised across architectures, scale, and a causal test; (2) the Level-2 co-expression ceiling, established with size-matched nulls; and (3) contextualisation as a model-characterisation metric that distinguishes tokenisation schemes (§4.7).

2. Related work

Contextual representations and word sense. Our methods descend directly from the NLP interpretability literature on contextualised embeddings. Ethayarajh (Ethayarajh 2019) showed that representations are anisotropic in every layer, that same-word self-similarity falls in upper layers (they become more context-specific), and that a static vector explains little of a word’s contextual variance — and, crucially, that every cosine must be corrected against an anisotropy baseline. Timkey and van Schijndel (Timkey and Schijndel 2021) showed that a few rogue dimensions dominate similarity and that a scalar correction is insufficient; Mu et al. (Mu et al. 2018) and Cai et al. (Cai et al. 2021) give the standard postprocessing and the local-isotropy picture. Coenen et al. (Coenen et al. 2019) and Garí Soler and Apidianaki (Soler and Apidianaki 2021) show word senses form separable clusters and that polysemy level is itself readable. We import both the measurements and the controls.

Emergent, causally-used latent variables. That a model represents and *uses* structure it was never given is established for board games (Li et al. 2023; Nanda et al. 2023; Karvonen 2024) and for physical coordinates in language models (Gurnee and Tegmark 2024). We adopt this literature’s evidentiary standard — a probe shows only correlation, so a causal claim requires an intervention against a matched control — and its steering machinery (Turner et al. 2023; Zou et al. 2023), of which the extraction and amplification of interpretable feature directions in production models (Templeton et al. 2024) is the large-scale instance. Our causal result is a steer-here/read-there test in this mould.

Single-cell foundation models and their representations. Geneformer (Theodoris et al. 2023), scGPT (Cui et al. 2024), scFoundation (Hao et al. 2024), scBERT (Yang et al. 2022), UCE (Rosen et al. 2023) and STATE (Adduri et al. 2025) differ in tokenisation (rank-ordered gene identity, value-binned expression, or protein-sequence-initialised gene tokens), and the review of Bian et al. (Bian et al. 2024) defines the “contextual gene embedding” our work dissects. Max-

Toki (Ortega et al. 2026), the primary model here, is a Llama-architecture, Geneformer-lineage model at 217M and 1B scales. scGPT reports cell-state-specific gene networks via attention, the closest prior to a “context changes the gene” claim — but operationalised as a pairwise network, never as a property of the representation vector, and without a decomposition or a null. The *static* gene-embedding geometry of these models has been mapped — its spectral structure (Kendiukhov 2026b), the topological and geometric structure it does and does not learn (Kendiukhov 2026g), and a decomposition of what a protein language model already supplies versus what single-cell pretraining adds (Kendiukhov 2026f) — but the *contextual* representation, the object here, has not.

What SCFMs do not capture. A critical literature finds SCFMs often fail to beat simple baselines: on zero-shot tasks (Kedzierska et al. 2025), under deeper evaluation (Boiarsky et al. 2024), on perturbation prediction (Ahlmann-Eltze et al. 2025; Bendidi et al. 2024), and across utilities (Liu et al. 2026); and sparse autoencoders find organised biological knowledge but minimal regulatory logic in these models (Kendiukhov 2026d), while SCFM attention re-expresses co-expression rather than unique regulatory signal (Kendiukhov 2026e). Our co-expression ceiling is the representation-geometry instance of this pattern, established with a matched null.

The biology of the ceiling. Co-expression is the canonical guilt-by-association signal (Dam S et al. 2018), and gene-embedding methods from Gene2vec (Du et al. 2019) to GenePT (YT and J 2023) are built on it. Genuine context-dependent gene function exists — moonlighting proteins (CJ 1999), lineage-determining transcription factors that select different enhancers with different partners by cell type (S et al. 2010) — but much of it is post-transcriptional and so, we argue, structurally invisible to an expression-only model, which is why our Level-2 tests are informative negatives.

3. Methods

Extraction. We run cells from Tabula Sapiens (immune, kidney, lung (Tabula Sapiens Consortium 2022)) through a model and record, for each gene in a panel, its hidden state at that gene’s token position, averaged over cells of one cell type. The headline MaxToki-217M extraction covers 12 cell types $\times$ 1000 cells, a 6000-gene panel, taps at layers 0, 4, 8, 11. Four design choices, each fixing a measured failure of a pilot: (1) **pairwise contexts**, not the all-context intersection, which leaves ~ 65 housekeeping genes — the “stopwords” that are *most* context-sensitive (Ethayarajh 2019) — whereas pairwise comparison recovers $\sim 5,700$ genes per pair; (2) **cell-level split halves**, each per-gene mean estimated twice from disjoint cells for an honest noise floor; (3) an **occurrence cap** (50 tokens per gene, context, partition), because a per-gene mean has standard error $1/\sqrt{n}$ and unequal counts inject context-dependent noise into any interaction term; (4) **per-dimension z-scoring** before any cosine, to remove the rogue-dimension inflation a scalar anisotropy correction misses (Timkey and Schijndel 2021).

Decomposition. For gene g in cell type c , write $v(g,c) = a(g) + b(c) + E(g,c)$: gene main effect $a(g)$ (identity), context main effect $b(c)$ (the **averaging null** — attention pools over the cell, so everything drifts together), and gene-specific interaction $E(g,c)$, the only term where biology can live.

Contextualisation strength (EXCESS). For a cell-type pair (c, c') , the *crowd-removed shift* of gene g is $\Delta(g) = [v(g,c) - v(g,c')] - \text{mean}_g [v(g,c) - v(g,c')]$ — the gene’s own shift minus the average shift over all genes (the “crowd”, i.e. the context main effect). We compute $\Delta(g)$ twice, from two disjoint halves of the cells (c, c') , and set $\text{EXCESS} = \text{mean}_g \cos(\Delta(g), \Delta(g)) - \text{mean}_g \cos(\Delta(g), \Delta(\text{perm } g))$: the same gene’s shift agreeing across independent cells, minus a gene-shuffled control. $\text{EXCESS} = 0$ means pure averaging (no gene-specific context response); it is averaged over all cell-type pairs.

Functional axes and nulls. A functional axis u is the unit vector separating two gene classes (from gene ontology) in the gene main effect: $u = \text{mean}(a(g)|\text{class A}) - \text{mean}(a(g)|\text{class B})$. Its **reproducible interaction power** is: project each gene’s two-way interaction term $E(g,c)$ onto u

to get a scalar per (gene, context), then average the product of the two cell-partitions’ scalars over count-balanced (gene, context) cells — a covariance that survives only if the modulation reproduces across independent cells. We compare this power for u against (i) matched **random-axis** nulls (random gene groupings of the same pole sizes); (ii) **size- and co-expression-matched** null gene-set pairs — the decisive control, since functional gene sets are co-expressed — where co-expression coherence is the mean pairwise Pearson correlation of expression across the extraction cells within the gene set, the null pairs are drawn to match the functional poles on both size and coherence, and the functional axis’s power is reported as an empirical percentile among them (skew-robust, because the null power distribution is right-skewed); and (iii) a **representation-tightness-matched** null (tightness = mean cosine among the axis genes’ own representations). Full parameters in §7.

4. Results

Sections 4.1–4.3, 4.5 and 4.7–4.8 establish **Level 1** (the model-fact: contextualisation is real, not position, replicable, causal, learned, and scaling); §4.4 and §4.6 test **Level 2** (novel biology) and find the co-expression ceiling.

4.1 Genes are contextualised gene-specifically, and it peaks in the early-middle layers

The gene-specific context response is large and highly reproducible (Table 1, Fig. 1). At layer 4 the gene-specific EXCESS is **+0.758**: the same gene’s crowd-removed shift agrees at cosine +0.759 across independent cells, while a *different* gene’s agrees at **+0.001** — the effect is entirely gene-specific. The context main effect replicates at +0.996, confirming the measurement works. Layer 0 (the context-free embedding output) returns **exactly 0.000**, the sanity check that the pipeline reports nothing when there is nothing. A scan across layers 0, 4, 8 and 11 places the peak at layer 4, rising through the early layers and decaying toward the output.

Layer	EXCESS (same – diff)	same-gene	diff-gene	main-effect replication
0 (embedding)	+0.000	+0.000	+0.000	—
4	+0.758	+0.759	+0.001	+0.996
8	+0.665	+0.666	+0.001	+0.996
11	+0.627	+0.628	+0.001	+0.997

Table 1. Gene-specific context response by layer (MaxToki-217M). 66 cell-type pairs; ~5,700 genes are shared per pair, of which a median ~1,700 remain after the occurrence-cap count-balancing (§3); bootstrap CIs ± 0.002 .

Metric validity (anisotropy). These representations are anisotropic, as expected (Ethayarajh 2019): the mean cosine of random different-gene pairs is +0.45 to +0.53 at layers 4–11 (raw), and same-gene self-similarity falls with depth (+0.94 $\rightarrow$ +0.82 $\rightarrow$ +0.75) — genes become more context-specific in upper layers, the single-cell echo of Ethayarajh’s finding. Our per-dimension z-scoring drives anisotropy to ~0.00 while keeping genes distinguishable (corrected self-similarity +0.89/+0.65/+0.53), and reduces the top-PC share (0.21 $\rightarrow$ 0.12 at layer 11), so the cosine-based metrics are computed in a corrected space.

4.2 It is not the gene’s rank position

MaxToki reads a cell as a rank-ordered gene list and encodes token position, so a gene ranking 5th in one cell type and 500th in another is an obvious confound. It is not the driver (Fig. 2). The gene-specific response is essentially identical for rank-stable genes (+0.759) and rank-moving genes (+0.755), and **96%** survives regressing the projection on per-(gene, context) mean rank (+0.728 residualised at layer 4). Shift *magnitude* correlates with rank change (-0.35), but its *direction* is largely orthogonal to rank.

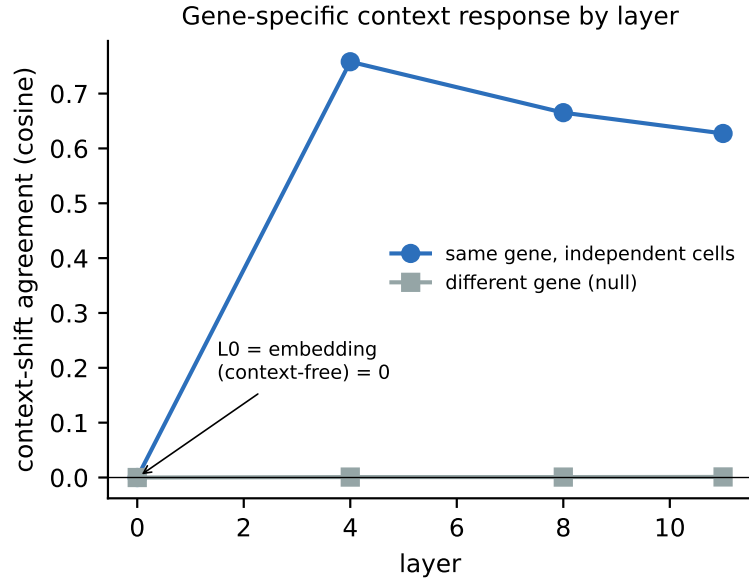


Figure 1: **Figure 1.** Gene-specific context response by layer. Same-gene agreement across independent cells (blue) far exceeds the gene-shuffled null (grey); layer 0, the context-free embedding, is exactly zero.

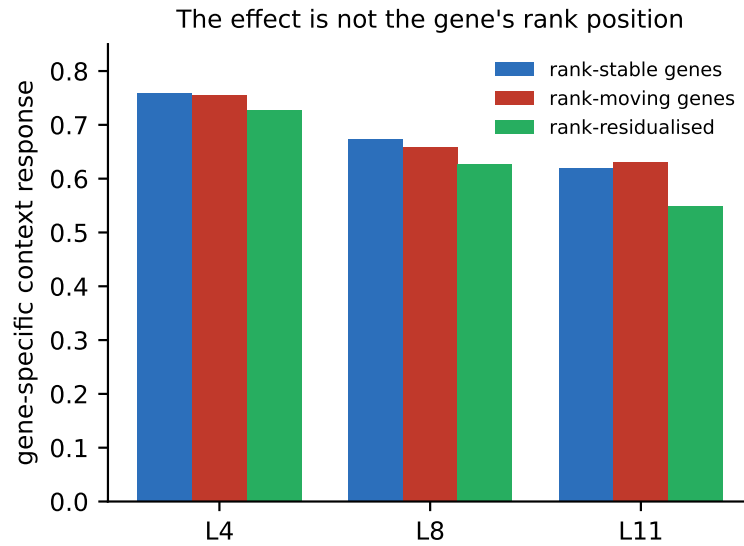


Figure 2: **Figure 2.** The effect is not rank position: gene-specific response is identical for rank-stable and rank-moving genes and survives residualising on mean rank.

4.3 It replicates on an independent dataset

All of the above is on one atlas. On an independent dataset — Setty CD34+ bone-marrow hematopoiesis (the same lineage in which a recoverable manifold was found in scGPT (Kendiukhov 2026a)), with eight *developmental states* rather than adult cell types as the context axis — the phenomenon replicates: EXCESS **+0.640** (layer 2) and **+0.584** (layer 4), gene-shuffled null +0.004, functional-z **+9.1/+7.6**. Neither the contextualisation nor its functional organisation is a Tabula-Sapiens or adult-cell-type artefact.

4.4 The modulation is functionally organised — but the organisation is co-expression

Context modulates genes along interpretable functional axes far more than along random directions. Reported as raw z (rank-controlled in parentheses), the nuclear/transcriptional vs. surface/secreted axis reaches $z = +21.9$ (+20.2); mitochondrial vs. cytoskeletal +10.3 (+8.3); transcription vs. transport +7.1 (+6.3); the abundance-neutral but weakly-separable cell-cycle-vs-differentiation axis is null, +0.7 (−0.7) at AUC 0.54. The nuclear/surface axis is biologically valid — cell types order sensibly, lymphocytes at the nuclear pole (CD8 T +4.1, CD4 T +3.7, B +3.4) and secretory/surface cells at the other (alveolar type-2 −5.3, type-1 −3.7, macrophage −3.2).

The decisive control asks whether this exceeds **co-expression** (Fig. 3). Functional gene sets are co-regulated modules, so their genes co-vary across cells regardless of any “understanding” of function. The right null must match the functional poles on **both** their size and their co-expression: comparing a ~1,880-gene functional axis against small random modules is invalid, because smaller sets reach higher coherence and different power. We therefore build null gene-set pairs of the *same sizes* as the functional poles, matched on co-expression coherence, and — because the null power distribution is heavily right-skewed — report the functional axis’s power as an empirical percentile among them. The nuclear/surface axis’s power (5.32) sits at the **p = 0.05** border of size-matched co-expression modules (mean 2.22); the mitochondrial/cytoskeletal axis at $p = 0.16$ and the transcription/transport axis at $p = 0.10$ are not significant. So the functional organisation does *not* robustly exceed co-expression — the headline axis is at best marginal, the others indistinguishable. It does clear a representation-space **tightness**-matched null (mean cosine among the axis genes’ own representations) and, by construction, the random-axis null ($p < 0.001$), so it is not merely a compact-gene-set artefact — but co-expression is sufficient. This is the geometry instance of the field-wide pattern that SCFM re-express co-expression (Kendiukhov 2026e; Kedzierska et al. 2025).

4.5 The functional-context direction is causally used

Level 1 is a genuine model-fact, and it is *used*, not decorative (Fig. 4). In a steer-here/read-there test we add a functional direction u to a random half of a cell’s genes and read the model’s own output logits at the *other*, untouched half. Nuclear-gene logits rise toward $+u$ (specific effect **+0.160**) and fall toward $-u$ (**−0.112**), a signed swing of **+0.272** versus **−0.015** for a norm-matched random push, dose-dependently across three strengths and beyond the random control on 24 of 30 cells ($p = 7 \times 10^{-5}$). The functional-context direction propagates through attention and shifts the model’s predictions.

The effect is not specific to one axis or layer. It replicates for the same nuclear/surface direction injected at a deeper layer (layer 8: swing +0.566, 30/30 cells, $p = 9 \times 10^{-1}$, stronger than at layer 4) and for an independent functional axis (mitochondrial vs. cytoskeletal at layer 4: swing +0.321, $p = 0.02$). It is null for the weakest-separated axis we tested causally (transcription vs. transport, validity AUC 0.66: swing −0.226, $p = 1.0$ — in fact a reversal), consistent with a channel that exists for well-formed functional directions but not for every ontology contrast one can write down. Across the four causal configurations the p-values are raw; the two strongest (nuclear/surface at layers 4 and 8) survive Bonferroni correction for the four tests, while the mitochondrial axis ($p = 0.02$) becomes marginal.

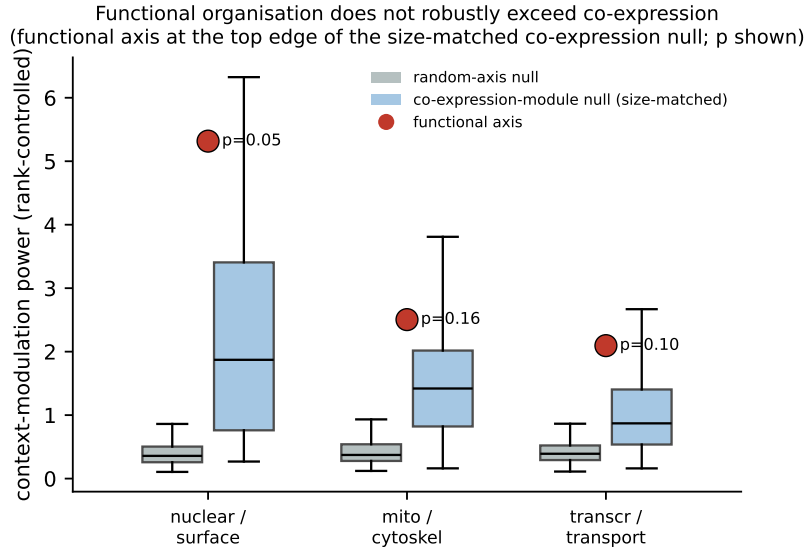


Figure 3: **Figure 3.** The crux control. Random gene modules (grey) trace context-modulation power against co-expression coherence; the functional axes (red) sit on that curve, so the functional organisation is co-expression.

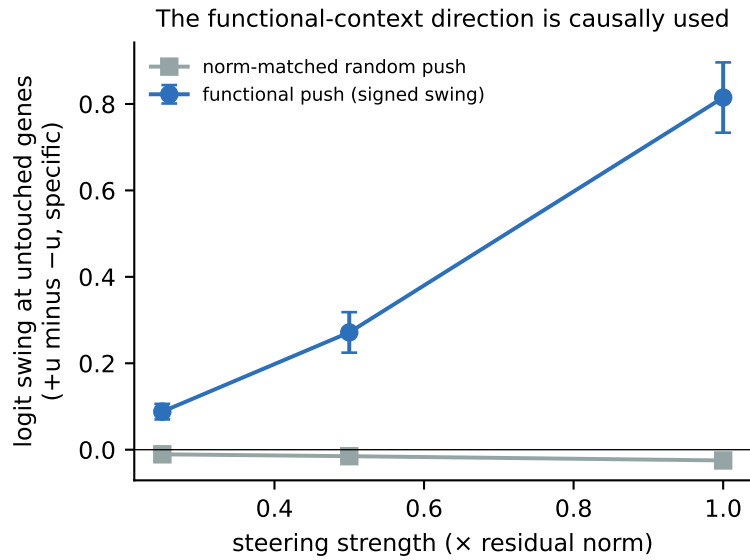


Figure 4: **Figure 4.** Causal use. Steering a functional direction into half a cell's genes shifts the model's logits at the *other* genes with the sign of the push, dose-dependently, far beyond a norm-matched random push.

4.6 No novel context-dependent gene function (the co-expression ceiling)

Two independent attempts to read *context-appropriate or novel* gene function beyond co-expression are null. (1) **Directional congruence** — does a gene move toward its own functional pole specifically in contexts sharing that character (a cross-validated non-additivity test)? Inside the co-expression null (empirical $p = 0.2$); no signal. (2) **Placement near curated targets** — are transcription factors positioned near their ChIP-seq / literature targets, context-appropriately (21 TFs)? Static excess $+0.023$ ($p = 0.19$), activity-modulation $= +0.080$ ($p = 0.33$); no signal, even with real target ground truth. A third attempt (curated context-switching TFs) is uninformative rather than null: only one such TF (RUNX1) is well-sampled in the panel, because context-switching TFs are typically low-expressed — itself an instance of the measurement wall. The boundary is consistent: an expression-only SCFM contextualises genes richly, but the information is co-expression, and does not resolve a gene’s changed role in a new cell beyond it (the ceiling is quantified in Fig. 3).

4.7 Contextualisation as a model metric: architecture, scale, and learned vs. architectural

The direction immune to the co-expression ceiling is to treat the phenomenon as a **model-quality metric** (Table 2, Fig. 5). Two findings. (1) **Contextualisation is universal and scales.** Every architecture strongly contextualises genes (EXCESS 0.70–0.88, main-effect replication > 0.93); on matched data (600 cells, identical settings) it increases with size, MaxToki-217M $+0.740 \rightarrow$ 1B $+0.881$, in line with the loss-scaling behaviour reported for these masked-reconstruction models (Kendiukhov 2026c). (2) **Architecture determines functional organisation.** Rank-based MaxToki organises context modulation along functional axes an order of magnitude more ($z = 16$ – 22) than value-binned scGPT ($+2.7$) or protein-sequence-initialised STATE ($+1.3$). STATE is telling: it moves genes strongly in context (EXCESS $+0.835$) but in no functionally coherent direction — its ESM2-initialised gene tokens carry sequence identity, not co-expression. The gap is real, not low power (pole sizes 366–1,390).

- (3) **The phenomenon is learned, not architectural.** An untrained, random-init MaxToki of identical architecture and vocabulary — the decisive control, since attention mixes tokens regardless of training — contextualises genes far less (EXCESS $+0.274$ vs. $+0.74$ – 0.76 trained) and organises that modulation functionally far less (functional- z $+3.8$ vs. $+21.2$). This separates two things cleanly. Contextualisation has a real architectural *floor* ($+0.274$: fixed random embeddings plus deterministic attention mixing give a reproducible, gene-specific shift on their own), but roughly two-thirds of the trained effect is learned. The functional organisation is *mostly* learned: the architecture alone produces a significant amount ($z = +3.8$), which training then amplifies $\sim 5.6\times$ (to $+21.2$) — so roughly 18% of the functional organisation is architectural and 82% is built by training, aligning context modulation with functional axes, and hence with co-expression.

Model	tokenisation	contexts	genes	dim	EXCESS	functional-z
scGPT	value-binned	8	5000	512	$+0.705$	$+2.7$
STATE-SE (state-embedding)	ESM2 gene tokens	9	1840	2048	$+0.835$	$+1.3$
MaxToki-217M	rank-based	12	5000	1232	$+0.740$	$+21.2$
MaxToki-1B	rank-based	12	5000	2304	$+0.881$	$+16.4$

Model	tokenisation	contexts	genes	dim	EXCESS	functional-z
MaxToki-217M (random init)	untrained control	12	6000	1232	+0.274	+3.8

Table 2. Cross-model comparison at layer 4. *EXCESS* = contextualisation strength; *functional-z* = organisation along the nuclear/surface axis vs. a random-axis null. *MaxToki-217M/1B* use matched 600-cell settings for a fair scaling comparison; the random-init row is the learned-vs-architectural control (§4.7); its 6000-gene panel vs. the 5000 of the trained rows does not affect *EXCESS*, which is a within-panel cosine statistic.

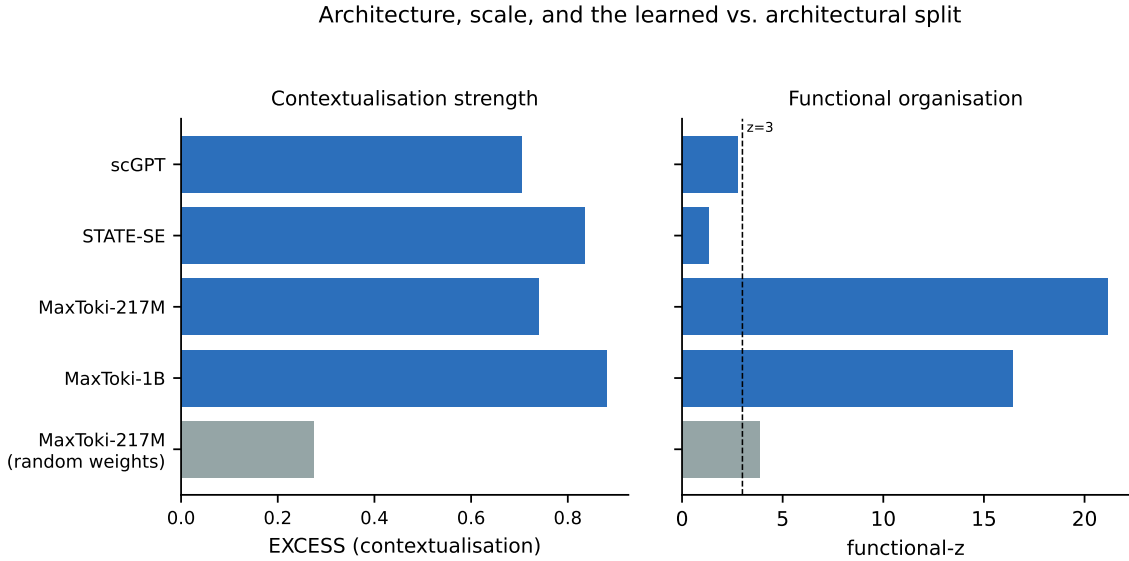


Figure 5: **Figure 5.** Contextualisation as a model metric across architecture, scale, and the random-weights control.

4.8 Context aids prediction, but not through the most-contextualised genes

If contextualisation served the autoregressive objective directly, the genes the model contextualises most should be the genes whose next-gene prediction most benefits from real context. It does not work that way. Using a chimeric-context intervention — the model’s log-probability of a target gene given its cell’s true rank-ordered predecessors versus a random other cell’s — real context helps prediction enormously (mean benefit **+7.7 nats**, a nat being a natural-log unit of log-probability; i.e. the true gene is roughly two-thousand-fold more probable under real than random context; 536 target genes). But the per-gene strength of that benefit does **not** track how much a gene is contextualised: the partial Spearman correlation, controlling for gene frequency, is **−0.087** (95% CI $[-0.168, -0.004]$) — at best faintly negative, and with per-gene estimates noisy on both sides, best read as *no positive link* rather than a firm negative one. Context is central to the model’s objective, and contextualisation is real, but the two are not aligned per gene: contextualisation is not simply the model encoding its per-gene predictive reliance on context.

5. Discussion

The result is a characterisation of *how* SCFMs represent genes in context, and a proof of *what* that representation carries. Positively: SCFMs reshape gene representations gene-specifically, reproducibly, causally, more with scale, and — for rank-based tokenisation — along co-expression-

aligned functional axes. This is, to our knowledge, the first measurement for single-cell models of the “how contextual are contextualised representations” question (Ethayarajh 2019), and the architecture/scale comparison is a usable interpretability metric that discriminates tokenisation schemes a downstream benchmark would not.

Negatively, and equally a contribution: the modulation does not carry a gene’s context-specific function beyond co-expression. The exciting version — reading a gene’s changed role in a new cell to generate hypotheses — is not reachable for an expression-only model by either route we tried, each with a matched-null control. Why the ceiling exists is mechanistically coherent: the model’s only inputs are which genes are co-expressed, so the richest thing its contextual geometry can encode about a gene is its co-expression neighbourhood; genuinely novel context-dependent function is frequently post-transcriptional (CJ 1999; S et al. 2010) and leaves no expression signature to latch onto. The finding therefore predicts where headroom is: multi-omic or sequence-aware models, not larger expression-only ones, are what a Level-2 result would require.

6. Limitations

Cross-architecture EXCESS is confounded by occurrence cap (20 for scGPT/STATE vs. 50 for MaxToki) and dimensionality (higher-dimensional models show higher EXCESS), so the 217M→1B increment reflects size *and* width, and even the cap-matched scaling pair still differs in width (dim 1232 vs. 2304) — the width-robust comparison is the functional-z architecture gap, which has ample power. The functional-z gap itself is measured on non-identical context sets across models (12 vs. 9 vs. 8 cell types); we did not re-run it on a common context set, so a residual context-composition confound remains. The causal test (§4.5) compares the functional direction only against a norm-matched *random* push, not against co-expression-coherence-matched directions, so it shows the direction is used, not that its causal use exceeds co-expression. STATE is evaluated on immune cells only in this study, so its cross-architecture point rests on a narrower panel than the others. The co-expression ceiling is a null result: the headline axis sits at the $p = 0.05$ border of the size-matched co-expression null and the other axes are non-significant, which shows the organisation does not clearly exceed co-expression, not that the two are provably identical; the empirical p depends on the null-module construction. The Level-2 null is evidence of a ceiling, not proof of absence — some context-dependent function is structurally invisible to an expression-only model, so those tests are weak by construction (pre-registered). Functional axes are built from one ontology file; the verdict is robust across three axes but specific z-scores depend on pole definitions.

7. Conclusions

Single-cell foundation models build contextual gene representations that are a genuine object of study: a gene’s vector is reshaped by its cellular context in a gene-specific, reproducible, learned, and causally used way that grows with model scale and is shaped by tokenisation. This is the first systematic characterisation of the phenomenon for single-cell models, and it yields a usable model metric — contextualisation strength and its functional organisation — that distinguishes architectures a downstream benchmark would not. At the same time, matched-null controls place a firm boundary on what that contextualisation carries: it does not robustly exceed co-expression, and two independent probes for novel context-dependent gene function are null. The practical implication is that reading a gene’s changed role in a new cell type — the application that would make these representations a hypothesis-generation tool — is not reachable with an expression-only model, and will require multi-omic or sequence-aware models rather than merely larger expression-trained ones.

8. Methods details and reproducibility

All analyses are deterministic (seed 0) and run on cached extractions. Extraction: `ctx_extract_maxtoki.py` (+ `_scgpt`, `_state`, `_devel`, `_random`). Analyses: `ctx_polysemy.py` (§4.1), `ctx_anisotropy.py`

(§4.1), `ctx_layer_curve.py` (§4.1), `ctx_position_confound.py` (§4.2), `ctx_independent.py` (§4.3), `ctx_functional_axes.py`, `ctx_coexpr_null_v2.py`, `ctx_tightness_null.py` (§4.4), `ctx_causal.py` across axes/layers (§4.5), `ctx_directional_probe.py` / `ctx_curated_targets.py` (§4.6), `ctx_cross_model.py` (§4.7), `ctx_prediction_link.py` (§4.8). Metric definitions: EXCESS as in §3; reproducible interaction power = mean over count-balanced (gene, context) of the product of the two partitions’ two-way interaction terms along a unit axis; functional-z, its z-score against 200–300 random-partition axes; co-expression coherence = mean within-set expression correlation. The co-expression null (`ctx_coexpr_null_v2.py`) draws null gene-set pairs of the *same sizes* as the functional poles, matched on coherence, and reports the functional axis’s power as an empirical percentile among them (skew-robust). Result artefacts are the corresponding `results/ctx_*.json`.

CRediT authorship contribution statement

Ihor Kendiukhov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All analysis code is openly available at <https://github.com/Biodyn-AI/gene-context>, including every analysis script, the intermediate result artefacts (`results/ctx_*.json`), the figure-generation code, and this manuscript with its bibliography. The study uses only publicly available models and data: the MaxToki-217M/1B checkpoints (Hugging Face, `theodoris-lab/MaxToki`); scGPT, Geneformer and STATE (public releases); the Tabula Sapiens atlas (Tabula Sapiens Consortium 2022) (CZ CELLxGENE); the Setty CD34+ bone-marrow dataset; and Gene Ontology annotations. The per-gene contextual representation tensors extracted from these models (the intermediate `.npz` files, several gigabytes) are not deposited because of their size, but are fully regenerable from the public inputs with the provided extraction scripts, and are available from the author on reasonable request. No new experimental data were generated.

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